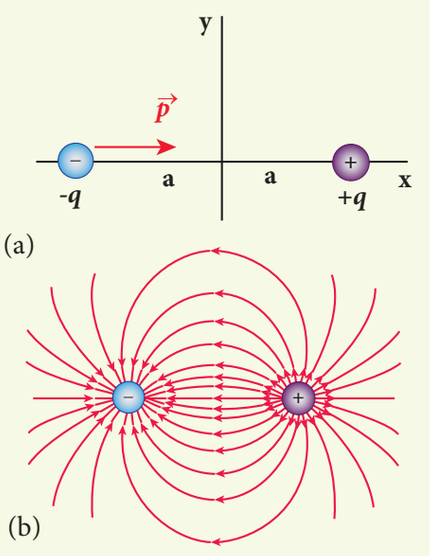


Figure 1.16 (a) Electric dipole (b) Electric field lines for the electric dipole

$$\vec{p} = qa\hat{i} \; -qa\left(-\hat{i}\right) = 2qa\hat{i} \qquad (1.11)$$

The electric dipole moment vector lies along the line joining two charges and is directed from –q to +q. The SI unit of dipole moment is coulomb meter (Cm). The electric field lines for an electric dipole are shown in Figure 1.16 (b).

- For simplicity, the two charges are placed on the x-axis. Even if the two charges are placed on y or z-axies, dipole moment will point from –q to +q. The magnitude of the electric dipole moment is equal to the product of the magnitude of one of the charges and the distance between them, $|\vec{p}| = 2qa$
- Though the electric dipole moment for two equal and opposite charges is defined, it is very general. It is possible to define and calculate the electric dipole moment for a single charge, two positive charges, two negative charges and also for more than two charges.

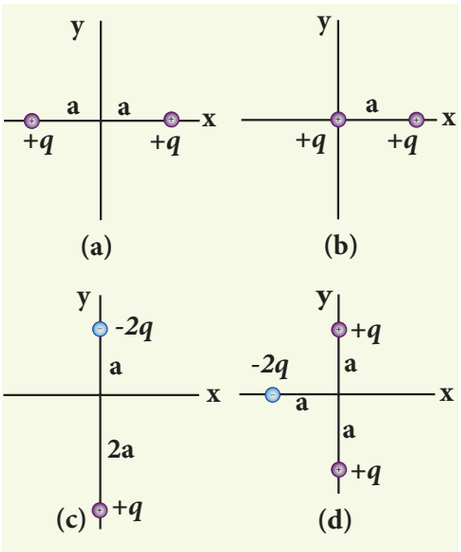
  For a collection of n point charges, the electric dipole moment is defined as follows:

$$\vec{p} = \sum_{i=1}^{n} q_i \vec{r}_i \qquad (1.12)$$

where $\vec{r}_i$ is the position vector of charge q_i from the origin.

EXAMPLE 1.10

Calculate the electric dipole moment for the following charge configurations.



Solution

Case (a) The position vector for the +q on the positive x-axis is $a\hat{i}$ and position vector for the +q charge the negative x axis is $-a\hat{i}$. So the dipole moment is,

$$\vec{p} = (+q)(a\hat{i}) + (+q)(-a\hat{i}) = 0$$

Case (b) In this case one charge is placed at the origin, so its position vector is zero. Hence only the second charge +q with position vector $a\hat{i}$ contributes to the dipole moment, which is $\vec{p} = qa\hat{i}$.

From both cases (a) and (b), we can infer that in general the electric dipole moment depends on the choice of the origin and charge configuration. But for one special case, the electric dipole moment is independent of the origin. If the total